

Shor's Algorithm, Period Finding and Discrete Logarithms

The abelian hidden subgroup problem: framework, two instances, the interference audit, and what it costs to build

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What this document is

A consolidation of a working conversation on Shor's algorithm together with three complementary sources: the general machinery of Childs, *Lecture Notes on Quantum Algorithms* (ch. 4–5); the group-algebra and number-theoretic treatment of Pittet, *Mathematical aspects of Shor's algorithm*, after Batty–Braunstein–Duncan–Rees; and the step-by-step walkthrough of Nguyen, Petruescu, Jadhav and Wong, from which the fully worked instance of §8 is drawn. Part I develops the abelian Fourier transform, phase estimation and the hidden subgroup problem in general, settling on the way what happens when the work register is an eigenvector, a superposition of eigenvectors, or a generic state (§4); Part II specialises to the two instances that break deployed cryptography; Part III is an *interference audit* — the exact law governing when a circuit's leftover data destroys the algorithm; Part IV is what it takes to build the thing. A worked instance ($N = 15$) is carried through in §8, where the abstract statements can be checked by eye. Companion reference sheets exist for the two instances separately; this document is the connective tissue between them. Numerical checks quoted throughout were run directly.

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Part I — The general machinery

1 Characters and the abelian Fourier transform

Let G be a finite abelian group. A *character* is a homomorphism $\omega : G \rightarrow \mathbb{C}^\times$; finiteness forces $|\omega(x)| = 1$, since $\omega(x)^{|G|} = \omega(x^{|G|}) = 1$. The characters form a group $\hat{G}$ under pointwise multiplication, and $\hat{\hat{G}} \cong G$ (non-canonically — an isomorphism requires choosing generators). The Fourier transform over G is

$$F_G := \frac{1}{\sqrt{|G|}} \sum_{x \in G} \sum_{y \in \hat{G}} \omega_y(x) |y\rangle\langle x|, \quad (1)$$

where ω_y is the y -th character. Unitarity is exactly the orthogonality of characters. Since $\hat{\hat{G}} \cong G$ we label characters by group elements throughout.

Why characters, and why they are one-dimensional. A character is the trace of an irreducible representation. For abelian G every irrep is one-dimensional — by Schur’s lemma, if G is abelian then each $\rho(g)$ commutes with all $\rho(h)$, hence is an intertwiner of the irrep with itself, hence scalar; and a representation in which every operator is scalar is irreducible only in dimension one. So for abelian groups “irrep” and “character” coincide, the Fourier transform is scalar-valued, and everything below goes through. This is precisely what fails in the nonabelian case (§14).

Two structural facts are used constantly.

Lemma 1 (Products factor). $\omega_{(y_1, y_2)}(x_1, x_2) = \omega_{y_1}(x_1) \omega_{y_2}(x_2)$, hence $\widehat{G_1 \times G_2} \cong \hat{G}_1 \times \hat{G}_2$ and

$$F_{G_1 \times G_2} = F_{G_1} \otimes F_{G_2}. \quad (2)$$

By the structure theorem every finite abelian group is a product of cyclic factors, so *the QFT over any finite abelian group is a tensor product of cyclic QFTs*. When a circuit shows two Fourier transforms side by side it is performing *one* transform, written in the coordinates the group came in.

Lemma 2 (Orthogonality). For $G = \mathbb{Z}_N$ with $\varepsilon_N := e^{2\pi i/N}$, $\sum_{x \in \mathbb{Z}_N} \varepsilon_N^{xd} = N[d \equiv 0 \pmod{N}]$.

Proof. If $d \equiv 0$ every term is 1. Otherwise $z = \varepsilon_N^d \neq 1$ satisfies $z^N = 1$, so $\sum_{x=0}^{N-1} z^x = (z^N - 1)/(z - 1) = 0$. $\square$

This one-line fact is the entire analytic content of every algorithm below. It is worth seeing it as a physical event rather than an identity: it says N complex amplitudes *cancel*, which requires that they be amplitudes of the same final state.

The simplest case is $G = \mathbb{Z}_2^n$, where $\omega_y(x) = (-1)^{x \cdot y}$ and

$$F_{\mathbb{Z}_2^n} = \frac{1}{\sqrt{2^n}} \sum_{x,y \in \mathbb{Z}_2^n} (-1)^{x \cdot y} |y\rangle\langle x| = H^{\otimes n}, \quad (3)$$

the transform behind Simon's problem: a layer of Hadamards is already a Fourier transform.

The register as a group algebra

There is a formulation that makes the appearance of characters inevitable rather than clever, and it is worth adopting because it explains a fact that is otherwise puzzling: why a layer of Hadamards and the $O(n^2)$ staircase of §2 are *both* Fourier transforms of the same register.

A single qubit is the group algebra of the two-element group, $V_1 = \mathbb{C}[\mathbb{Z}/2\mathbb{Z}] \cong \mathbb{C}^2$, with the group elements 0, 1 as the distinguished orthonormal basis. An n -qubit register is $V_n = V_1^{\otimes n}$, and the identifications

$$\mathbb{C}[\mathbb{Z}/2\mathbb{Z}]^{\otimes n} \xrightarrow{\sim} \mathbb{C}[(\mathbb{Z}/2\mathbb{Z})^n] \xrightarrow{\sim} \mathbb{C}[\mathbb{Z}/2^n\mathbb{Z}], \quad e_{i_1} \otimes \cdots \otimes e_{i_n} \mapsto (i_1, \dots, i_n) \mapsto \sum_k i_k 2^{k-1}, \quad (4)$$

are isomorphisms of Hilbert spaces, because each of the three families is orthonormal.

One Hilbert space, three groups. The register does not “have” a group; a group is something we *impose* on its basis labels. Choosing $(\mathbb{Z}/2\mathbb{Z})^n$ makes the characters $(-1)^{x \cdot y}$ and the Fourier transform is $H^{\otimes n}$. Choosing $\mathbb{Z}/2^n\mathbb{Z}$ makes the characters $\varepsilon_{2^n}^{xy}$ and the Fourier transform is the staircase of §2. *Same 2^n amplitudes, different notion of “periodic”, hence different algorithms:* Simon's problem lives in the first, Shor's in the second. The choice of group is the choice of what structure the algorithm is allowed to see.

This also gives the cleanest possible definition of the transform, free of matrices. In $L^2(\mathbb{Z}/N\mathbb{Z})$ with the delta basis $\{\delta_x\}$, the normalised characters $\{\chi_c/\sqrt{N}\}$ form a second orthonormal basis, and

$$F \text{ is the unique unitary with } \frac{\chi_c}{\sqrt{N}} \mapsto \delta_c \quad (5)$$

equivalently $F(\sum_x f(x)\delta_x) = \sum_c \hat{f}(c)\delta_c$ with $\hat{f}(c) = (f, \chi_c/\sqrt{N})$. *The Fourier transform is the change of basis from characters to deltas* — from “frequencies” to “positions”. Every algorithm below is an arrangement for making the answer a *position* in the basis one is about to measure.

Measurement, stated once and precisely

Since measurement is where all the probabilities enter, it is worth fixing rigorously. A *measurement* on a finite-dimensional Hilbert space V is a family of orthogonal projections $P_1, \dots, P_k$ ($P_i^2 = P_i = P_i^*$) with

$$P_i P_j = 0 \ (i \neq j), \quad \sum_{i=1}^k P_i = \text{id}_V. \quad (6)$$

Applied to a unit vector v , the outcome is i with probability $\mathbb{P}(i) = \|P_i v\|^2$, and the post-measurement state is $P_i v / \|P_i v\|$. Normalisation is Pythagoras: $\sum_i \|P_i v\|^2 = \|\sum_i P_i v\|^2 = \|v\|^2 = 1$.

Two instances are used constantly below. Measuring only the second of two registers is the family $\{\text{id} \otimes P_b\}_b$ with P_b the projection onto the line $\mathbb{C}b$; measuring only the first is $\{P_c \otimes \text{id}\}_c$. Both are genuine measurements in the sense above, which is exactly why partial measurement obeys the squared-norm rule used in §4 and §10 rather than the modulus-squared-of-an-amplitude rule.

2 The QFT over $\mathbb{Z}_{2^n}$

For $G = \mathbb{Z}_{2^n}$ the transform admits a product form that gives the circuit directly. Writing $x = x_{n-1} \dots x_1 x_0$ and $y = \sum_k y_k 2^k$,

$$F_{\mathbb{Z}_{2^n}} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \mathbb{Z}_{2^n}} \varepsilon_{2^n}^{xy} |y\rangle = \frac{1}{\sqrt{2^n}} \sum_y \prod_{k=0}^{n-1} \varepsilon_{2^n}^{x y_k 2^k} |y_{n-1} \dots y_0\rangle = \bigotimes_{k=0}^{n-1} |z_k\rangle, \quad (7)$$

$$|z_k\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + \varepsilon_{2^{n-k}}^x |1\rangle \right) = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i (x_0 2^{k-n} + x_1 2^{k+1-n} + \dots + x_{n-1-k} 2^{-1})} |1\rangle \right). \quad (8)$$

So $F|x\rangle$ is a *product* state, and the k -th output qubit depends only on the $n - k$ least significant bits of x . That is exactly a staircase of Hadamards and controlled phase rotations $R_k = \text{diag}(1, \varepsilon_{2^k})$, using $O(n^2)$ gates. Dropping the rotations with $k = \Omega(\log n)$ — angles too small to matter — gives an $O(n \log n)$ circuit accurate to $1/\text{poly}(n)$.

Remark. The product form is also why the QFT is never the bottleneck. In both algorithms below the arithmetic costs $\Theta(n^3)$ while the transform costs $O(n^2)$ or less.

3 Phase estimation

Given a unitary U (as a circuit, or as a black box supplying controlled- U^j) and an eigenvector $U|\psi\rangle = e^{i\phi}|\psi\rangle$, estimate ϕ . Prepare $2^{-n/2} \sum_{x \in \mathbb{Z}_{2^n}} |x, \psi\rangle$, apply $\sum_x |x\rangle\langle x| \otimes U^x$ to obtain

$$\frac{1}{\sqrt{2^n}} \sum_{x \in \mathbb{Z}_{2^n}} e^{i\phi x} |x, \psi\rangle, \quad (9)$$

then inverse-Fourier the first register and measure.

Two observations make this work, and both are worth stating explicitly.

(i) *Controlled powers cost log, not linear.* Writing $x = \sum_m x_m 2^m$, the controlled powers act as $|x\rangle|\psi\rangle \mapsto |x\rangle U^x |\psi\rangle$ because the m -th gate applies U^{2^m} exactly when $x_m = 1$ and the gates share one target, so their actions *compose*: $\prod_m U^{x_m 2^m} = U^{\sum_m x_m 2^m}$. Multiplying operators adds exponents, which is precisely how the binary digits of x reassemble the integer. No tensor product of the U^{2^m} appears anywhere; $\otimes$ ranges only over control qubits.

(ii) *The control must be controlled.* Applying U^x uncontrolled multiplies the whole state by a *global* phase, which is unobservable. Only the superposition over x , with a different power on each branch, converts the eigenvalue into *relative* phases. Equation (9) is a *phase ramp*: uniform moduli, phase advancing linearly in x . And a linear ramp of slope $s/2^n$ is exactly the Fourier basis vector $F_{2^n}|s\rangle$ — so reading the slope is the inverse transform.

If $\phi = 2\pi y/2^n$ exactly, the state (9) is $F_{2^n}|y\rangle \otimes |\psi\rangle$ and the outcome is certain. In general the outcome law is

$$\Pr(y) = \frac{1}{2^{2n}} \cdot \frac{\sin^2(2^{n-1}\phi)}{\sin^2(\frac{\phi}{2} - \frac{\pi y}{2^n})}. \quad (10)$$

Radians versus cycles. Writing phases in cycles, $\phi = 2\pi\theta$, $Q = 2^n$, and $\delta := \theta - y/Q$, equation (10) becomes the normalised squared Dirichlet kernel

$$|A_y(\theta)|^2 = \frac{\sin^2(\pi Q \delta)}{Q^2 \sin^2(\pi \delta)}, \quad A_y(\theta) = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi i x(\theta - y/Q)}. \quad (11)$$

The two are identical: $\frac{\phi}{2} - \frac{\pi y}{Q} = \pi \delta$, and $\sin^2(2^{n-1}\phi) = \sin^2(\pi Q \theta) = \sin^2(\pi Q \delta)$ because y is an integer and $\sin^2$ has period π . (Verified numerically to machine precision.) Cycles are the more convenient convention downstream, and are used from here on.

Lemma 3 (Sharpness). *Let y^* be the integer nearest $Q\theta$, so $|\delta| \leq 1/(2Q)$. Then $|A_{y^*}(\theta)|^2 \geq 4/\pi^2 \approx 0.405$.*

Proof. For $|u| \leq \pi/2$ concavity gives $|\sin u| \geq \frac{2}{\pi}|u|$; apply it with $u = \pi Q\delta$ to bound the numerator by $2Q|\delta|$, and $|\sin(\pi\delta)| \leq \pi|\delta|$ to bound the denominator. Then $|A_{y^*}| \geq \frac{1}{Q} \cdot \frac{2Q|\delta|}{\pi|\delta|} = \frac{2}{\pi}$. $\square$

4 Phase kickback, and what the work register must be

Section 3 assumed the work register holds an eigenvector. That assumption is never satisfied in Shor’s algorithm, and understanding exactly how it fails — and why the algorithm survives — is the conceptual heart of the subject. This section treats the three possibilities uniformly: eigenvector, superposition of eigenvectors, generic vector. All three are the *same state* analysed with different inputs, and one theorem covers them all.

The one state everything comes from

Fix a unitary U on a D -dimensional work space, a control register of t qubits, $Q = 2^t$, and any work state $|\phi\rangle$. After the Hadamards and the controlled powers,

$$|\Psi_\phi\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle \otimes U^x |\phi\rangle \quad (12)$$

Nothing else will be assumed. Cases 1–3 below are (12) for three choices of $|\phi\rangle$, and the differences between them are differences in a single object: the family of *branch states* $U^x |\phi\rangle$ and their mutual overlaps.

Phase kickback: the minimal statement

Proposition 4 (Kickback). *Let $U|u\rangle = e^{2\pi i\theta}|u\rangle$. Then for any control amplitudes α, β ,*

$$\text{c-}U[(\alpha|0\rangle + \beta|1\rangle) \otimes |u\rangle] = (\alpha|0\rangle + \beta e^{2\pi i\theta}|1\rangle) \otimes |u\rangle. \quad (13)$$

Proof. By definition $\text{c-}U = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes U$, so the state maps to $\alpha|0\rangle \otimes |u\rangle + \beta|1\rangle \otimes U|u\rangle = \alpha|0\rangle \otimes |u\rangle + \beta|1\rangle \otimes e^{2\pi i\theta}|u\rangle$. Move the scalar across the tensor sign — legitimate because $\otimes$ is bilinear, $|a\rangle \otimes (\lambda|b\rangle) = (\lambda|a\rangle) \otimes |b\rangle = \lambda(|a\rangle \otimes |b\rangle)$ — and factor out $|u\rangle$. $\square$

The proof is three lines and consists entirely of moving a scalar across a tensor sign. Nothing physically travelled from target to control; what changed is which description is useful. A *scalar has no address*: it belongs to the whole branch, not to either factor. Three things happen at once, and they are the crux.

- (i) The target does not move.** An eigenvector is a fixed point of U up to a scalar, so the controlled gate *cannot* change the work register; it can only emit a number. The work register acts as a catalyst — a phase source. Given no-cloning, this is what allows the single copy of $|u\rangle$ to be reused by all t controlled blocks (Case 1 below).
- (ii) The scalar lands on the control.** Branch $|0\rangle$ acquires 1 and branch $|1\rangle$ acquires $e^{2\pi i\theta}$: what would have been an unobservable *global* phase has become a *relative* phase between control basis states, which is physical and interferable. Information has flowed *backwards*, opposite to the way the gate “points”.
- (iii) The registers stay a product.** No entanglement is created. All the information now sits in the control alone, and the work register may be ignored from here on.

Point (ii) explains why phase estimation cannot be built from *uncontrolled* applications of U , no matter how many: $U^x|u\rangle = e^{2\pi i x \theta}|u\rangle$ is a global phase and is invisible to every measurement. The control is not a convenience; it is the mechanism that converts an unobservable number into an observable one.

Point (iii) deserves emphasis because it is widely misremembered. **Phase kickback is not an entangling phenomenon.** It is the statement that eigenvectors are precisely the states on which a controlled gate is coherence-preserving.

The master identity: branch overlaps

Apply $F_Q^\dagger \otimes \mathbb{I}$ to (12) and group by control value:

$$(F_Q^\dagger \otimes \mathbb{I})|\Psi_\phi\rangle = \sum_{y=0}^{Q-1} |y\rangle \otimes |w_y\rangle, \quad |w_y\rangle = \frac{1}{Q} \sum_x e^{-2\pi i xy/Q} U^x |\phi\rangle. \quad (14)$$

Only the control is measured, so the outcome probability is the *squared norm* of the leftover work vector, $\Pr[y] = \langle w_y | w_y \rangle$ — the same partial-measurement Born rule used in §10. Expanding, and using $\langle U^{x'} \phi | U^x \phi \rangle = \langle \phi | U^{x-x'} | \phi \rangle$,

Master identity: the outcome law is set by the branch overlaps

$$\Pr[y] = \frac{1}{Q^2} \sum_{x,x'} e^{-2\pi i (x-x')y/Q} \underbrace{\langle U^{x'} \phi | U^x \phi \rangle}_{= z_{x-x'}}, \quad z_\Delta := \langle \phi | U^\Delta | \phi \rangle. \quad (15)$$

Collecting terms by $\Delta = x - x'$, of which there are $Q - |\Delta|$,

$$\Pr[y] = \frac{1}{Q^2} \sum_{\Delta=-(Q-1)}^{Q-1} (Q - |\Delta|) e^{-2\pi i \Delta y/Q} z_\Delta, \quad (16)$$

the Fourier transform of the *autocorrelation* z_Δ of $|\phi\rangle$ under U , through a triangular window. Both forms verified numerically against direct simulation, to machine precision, in all three cases below.

The quantity z_Δ is exactly the object of Part III: it is the overlap of the work-register states left behind by two control branches. And it carries *two* pieces of information at once, which is the single most useful thing to know about this subject:

$\arg z_\Delta$	the signal — the kicked-back phase we want to read
$ z_\Delta $	the visibility — how much of the interference survives

An eigenvector gives $|z_\Delta| = 1$ with $\arg z_\Delta$ linear in Δ : maximal visibility, clean signal. Everything that goes wrong in Cases 2 and 3 is $|z_\Delta|$ falling below 1, i.e. branches leaving distinguishable records.

The spectral-measure theorem

Because U is unitary its eigenvectors $\{|u_k\rangle\}$ form an orthonormal basis, so *every* $|\phi\rangle$ expands as $|\phi\rangle = \sum_k c_k |u_k\rangle$ with $U|u_k\rangle = e^{2\pi i \theta_k} |u_k\rangle$. Define the *spectral measure* of $|\phi\rangle$ with respect to U ,

$$\mu_\phi := \sum_k |c_k|^2 \delta_{\theta_k}, \quad \text{a probability measure on } [0, 1). \quad (17)$$

Theorem 5 (Phase estimation is a spectrometer). *For any work state $|\phi\rangle$,*

$$\Pr[y] = \int_0^1 |A_y(\theta)|^2 d\mu_\phi(\theta) = \sum_k |c_k|^2 |A_y(\theta_k)|^2 \quad (18)$$

with A_y the Dirichlet amplitude of (11).

Proof. $z_\Delta = \langle \phi | U^\Delta | \phi \rangle = \sum_{k,k'} \bar{c}_{k'} c_k e^{2\pi i \Delta \theta_k} \langle u_{k'} | u_k \rangle = \sum_k |c_k|^2 e^{2\pi i \Delta \theta_k}$, the off-diagonal terms dying by orthonormality of the eigenbasis. Substituting into (15),

$$\Pr[y] = \sum_k |c_k|^2 \frac{1}{Q^2} \sum_{x,x'} e^{2\pi i (x-x')(\theta_k - y/Q)} = \sum_k |c_k|^2 \left| \frac{1}{Q} \sum_x e^{2\pi i x(\theta_k - y/Q)} \right|^2. \quad \square$$

This is the whole story in one line. **Phase estimation measures the spectral measure of the work state, blurred by the Dirichlet kernel.** It is a spectrometer: feed it a state, it reports which eigenfrequencies of U that state contains, with weights $|c_k|^2$ and resolution $1/Q$. The three cases are three shapes of μ_ϕ — a single spectral line, a few lines, or a broadband continuum — and nothing else distinguishes them.

Case 1: the work register is an eigenvector

Let $|\phi\rangle = |u\rangle$ with $U|u\rangle = e^{2\pi i \theta} |u\rangle$, so $\mu_\phi = \delta_\theta$. Applying Proposition 4 on every branch of (12):

$$|\Psi_u\rangle = \frac{1}{\sqrt{Q}} \sum_x |x\rangle \otimes e^{2\pi i x \theta} |u\rangle = \underbrace{\left(\frac{1}{\sqrt{Q}} \sum_x e^{2\pi i x \theta} |x\rangle \right)}_{|\Phi_\theta\rangle, \text{ the phase ramp}} \otimes |u\rangle. \quad (19)$$

Theorem 5 gives $\Pr[y] = |A_y(\theta)|^2$, so by Lemma 3 the nearest t -bit approximation to θ appears with probability $\geq 4/\pi^2$, and with certainty if θ is on the grid. Equivalently $z_\Delta = e^{2\pi i \Delta \theta}$: a pure tone, unit modulus for every Δ .

Entanglement: absent, and its absence is essential. Equation (19) is manifestly a *product*; the Schmidt rank is 1 (confirmed numerically). No step of the derivation uses entanglement, correlation, or measurement of the work register — only bilinearity. Two consequences follow, and both are needed later. (a) The work register emerges *unchanged*, so the single copy of $|u\rangle$ — which no-cloning forbids duplicating — survives to be reused by every controlled block in the circuit. (b) Discarding the work register costs nothing, because a product state has no correlations to lose.

Geometric reading: an arrow on the Bloch sphere. Take a single control qubit. It starts at $|+\rangle$, on the equator along $+x$. Proposition 4 rotates it by $2\pi\theta$ about the z -axis. *The eigenphase is literally an angle of rotation.* Measuring in the X basis gives $p(\pm) = \frac{1 \pm \cos 2\pi\theta}{2}$, which determines $\cos 2\pi\theta$; the Y basis gives the sine. The t -qubit version does the same thing t times at doubling angular speeds, which is binary place-value notation for an angle.

Quantum-optical reading: a phase plate. Read the control as the which-arm degree of freedom of an interferometer and the work register as a sample placed in one arm. An eigenvector is a sample that imparts a pure phase delay and *records nothing* about which arm the photon took. Fringes at full visibility; the phase shift is read off their position.

Case 2: the work register is a superposition of eigenvectors

Now $|\phi\rangle = \sum_k c_k |u_k\rangle$ with at least two nonzero c_k . Linearity of the circuit applies (19) to each eigencomponent *simultaneously*:

$$|\Psi_\phi\rangle = \frac{1}{\sqrt{Q}} \sum_x |x\rangle \otimes \sum_k c_k e^{2\pi i x \theta_k} |u_k\rangle = \sum_k c_k |\Phi_{\theta_k}\rangle \otimes |u_k\rangle, \quad (20)$$

after swapping the two finite sums. Theorem 5 then gives

$$\Pr[y] = \sum_k |c_k|^2 |A_y(\theta_k)|^2, \quad (21)$$

an *incoherent mixture* of the ideal single-eigenvector distributions. Operationally: *draw k with probability $|c_k|^2$, then run ideal phase estimation on θ_k .*

Entanglement: present, and here it is the mechanism — the derivation cannot be done without it. Equation (20) is a Schmidt decomposition with rank equal to the number of nonzero c_k (numerically: rank 2 for a two-component superposition, versus rank 1 in Case 1). Three points where the derivation depends on it:

(1) *Killing the cross terms.* In Theorem 5 the terms $k \neq k'$ vanish *only* because $\langle u_{k'} | u_k \rangle = \delta_{kk'}$. Were the eigenvectors merely linearly independent, $\Pr[y]$ would carry interference between different eigenbranches and the clean mixture would be false. Orthogonality of eigenvectors belonging to distinct eigenvalues (Lemma: $\langle u' | u \rangle = \langle u' | U^\dagger U | u \rangle = \bar{\lambda}' \lambda \langle u' | u \rangle$ with $\bar{\lambda}' \lambda \neq 1$) is thus load-bearing, not decorative.

(2) *The answer lives in the correlation.* The circuit has *not* computed all eigenvalues into one register. It has built a coherent correspondence $|\Phi_{\theta_k}\rangle \leftrightarrow |u_k\rangle$. Measuring the control therefore *conditions* the work register: observing y leaves it in the renormalised branch $\propto \sum_k c_k A_y(\theta_k) |u_k\rangle$, whose squared coefficients are precisely the posterior $\Pr[k | y]$. Collapse of one register is Bayesian updating of the other, carried out coherently.

(3) *One sample, not an average.* Because the branches carry perfectly distinguishing labels they cannot blur; the r ramps coexist and measurement *samples* one eigenphase rather than averaging them. This is an equality of statistics, not a hidden classical coin — the circuit stays coherent throughout.

Geometric reading: a shortened arrow. With one control qubit and $z = \langle \phi | U | \phi \rangle$, tracing out the work register gives

$$\rho_C = \frac{1}{2} \begin{pmatrix} 1 & \bar{z} \\ z & 1 \end{pmatrix}, \quad \vec{r} = (\text{Re } z, \text{Im } z, 0), \quad \text{Tr } \rho_C^2 = \frac{1 + |z|^2}{2}, \quad p(\pm) = \frac{1 \pm \text{Re } z}{2}. \quad (22)$$

So the control's Bloch vector lies in the equatorial plane; *its angle is the phase you want to read and its length is exactly $|z|$* . By Cauchy–Schwarz $|z| \leq 1$ with equality iff $U|\phi\rangle \propto |\phi\rangle$ — so $|z| = 1$, purity 1, product state and “eigenvector” are four names for one condition. A superposition of eigenvectors shrinks the arrow to $|z| < 1$: **entanglement with the work register shortens the very arrow whose angle you are trying to measure.** (Numerically: $|z| = 0.8358$ for a two-component superposition, with Bloch length and purity matching (22) to four decimals.) **Interferometric reading.** The sample now recoils differently depending on the arm, so it retains partial which-path information; fringe visibility falls to $|z|$, obeying the complementarity relation $V^2 + D^2 \leq 1$ with $D = \sqrt{1 - |z|^2}$.

Why this case is the one that matters

Case 2 is not a degradation to be avoided — it is **how Shor’s algorithm actually runs**. One cannot prepare $|u_k\rangle$ without knowing r , the very unknown; but $|1\rangle = r^{-1/2} \sum_k |\psi_k\rangle$ (Lemma 7) is the uniform superposition of *all* of them and costs one X gate. The circuit is linear, so running it on $|1\rangle$ runs it on every eigenbranch at once; the branches are orthogonal, so they cannot interfere; and measurement collapses onto one, delivering an estimate of s/r for a uniformly random s *not chosen by us*. That randomness is the price of not knowing the eigenvector, and the continued-fraction stage (§7) pays it. See §7 for the instance and §9 for its two-dimensional analogue.

Case 3: the work register is a generic vector

Mathematically there is nothing new: the eigenvectors of U span the space, so a generic $|\phi\rangle$ is a superposition of eigenvectors and Case 3 is a sub-case of Case 2. The difference is entirely quantitative, and it is decisive.

Take $|\phi\rangle$ Haar-random in dimension D . Then $|c_k|^2 \approx 1/D$ spread across all D eigenphases, so μ_ϕ is a broadband continuum rather than a few lines, and

$$z_\Delta = \sum_k |c_k|^2 e^{2\pi i \Delta \theta_k} \quad (23)$$

is a sum of D phasors with near-equal weights and incommensurate phases: it *decays*. Exactly, averaging over Haar-random $|\phi\rangle$,

$$\mathbb{E} |\langle \phi | U | \phi \rangle|^2 = \frac{D + |\text{Tr } U|^2}{D(D+1)} = O(1/D) \quad \text{whenever } |\text{Tr } U|^2 = O(D), \quad (24)$$

verified by sampling to three decimals at $D = 4, 16, 64, 256$. So $|z_\Delta| \sim D^{-1/2}$: the visibility, and with it the Bloch vector of (22), collapses toward zero.

By Theorem 5 the outcome distribution is D Dirichlet peaks superposed with weights $\approx 1/D$, which is near-uniform once $D \gtrsim Q$. Simulation at $Q = 32$, averaging over random U and $|\phi\rangle$:

	D	1 (eigenvector)	4	64	256
peak $\text{Pr}[y]$		0.7702	0.4562	0.0957	0.0644
mean $ z_1 $		1.0000	0.4814	0.1728	0.0912

Uniform would be $1/Q = 0.0312$. The peak decays toward it as the spectral measure spreads.

Entanglement: maximal, and information minimal. The Schmidt rank of (20) is now $\approx D$, so the control and work registers are close to maximally entangled and ρ_C is close to maximally mixed. **This is the inversion worth noticing:** across the three cases entanglement rises $0 \rightarrow \text{moderate} \rightarrow \text{maximal}$ while usefulness falls $\text{high} \rightarrow \text{high} \rightarrow \text{zero}$. Entanglement is therefore *not* the resource — what matters is the *structure* of the correlation: whether the branch overlaps z_Δ have unit modulus (Case 1), block structure with orthogonal labels and few atoms (Case 2), or decay (Case 3).

Note precisely what fails. The circuit is unitary and runs correctly; the measurement returns a legitimate sample from μ_ϕ . What is lost is the *inference*: with D unknown eigenphases and no structure relating them, one sample from a near-uniform distribution supports no conclusion. Case 3 is not a hardware failure but an *epistemic* one — and it is precisely the failure mode induced by uncleared scratch in Part III, where the work register’s accidental spreading was created by the circuit itself rather than by the choice of input.

The three cases side by side

	1. eigenvector	2. superposition	3. generic
$ \phi\rangle$	$ u\rangle$	$\sum_k c_k u_k\rangle$	Haar-random in $\mathbb{C}^D$
spectral measure μ_ϕ	δ_θ (one line)	few atoms	$\approx$ uniform on D atoms
overlap z_Δ	$e^{2\pi i \Delta \theta}, z = 1$	$\sum_k c_k ^2 e^{2\pi i \Delta \theta_k}$	decays, $ z_\Delta \sim D^{-1/2}$
state (12)	product, rank 1	entangled, rank $= \#\{c_k \neq 0\}$	entangled, rank $\approx D$
$\Pr[y]$	$ A_y(\theta) ^2$	$\sum_k c_k ^2 A_y(\theta_k) ^2$	near-uniform if $D \gtrsim Q$
Bloch length $ z $	1	< 1	$\rightarrow 0$
outcome	θ to t bits, $\geq 4/\pi^2$	one random θ_k , w.p. $ c_k ^2$	no usable information
work register after	unchanged, reusable	collapsed onto $ u_k\rangle$	scrambled
entanglement needed in the derivation?	no — bilinearity alone	yes — orthogonality kills the cross terms; the answer is the correlation	yes — and it is the obstruction

The moral

One state, (12); one theorem, (18); three shapes of spectrum. Phase kickback itself is elementary and entanglement-free — a scalar moved across a tensor sign. Entanglement appears the moment the work register is *not* an eigenvector, and whether that helps or hurts depends entirely on whether the resulting branch labels are *few and orthogonal* (Case 2: the correlation is the answer, and one measurement reads it) or *many and spread* (Case 3: the correlation is noise). Shor’s algorithm lives in Case 2 by necessity — $|1\rangle$ is preparable and $|\psi_k\rangle$ is not — and the entire remaining design effort, in Part III and Part IV, is spent keeping the circuit in Case 2 and out of Case 3.

5 The QFT over $\mathbb{Z}_N$ from the cyclic shift

The staircase circuit of §2 needs N a power of two. For general N there is an elegant route through phase estimation, and it closes a loop with the period-finding story below.

Consider the unitary that adds one modulo N — *the cyclic shift*:

$$U := \sum_{x \in \mathbb{Z}_N} |x+1\rangle\langle x|. \quad (25)$$

Its eigenvectors are exactly the Fourier basis states $|\tilde{x}\rangle := F_{\mathbb{Z}_N} |x\rangle$:

$$F_{\mathbb{Z}_N}^\dagger U F_{\mathbb{Z}_N} = \sum_{x \in \mathbb{Z}_N} \varepsilon_N^{-x} |x\rangle\langle x| \quad (26)$$

(verified numerically). So running phase estimation on U with $O(\log N)$ bits performs $|\tilde{x}, 0\rangle \mapsto |\tilde{x}, x\rangle$; running that in reverse erases the first register and leaves $F_{\mathbb{Z}_N}$ itself. Combined with Lemma 1, this gives the QFT over *any* finite abelian group.

The shift appears twice, in opposite roles

The operator $U|x\rangle = |x+1\rangle$ is the same object in both halves of the theory, and it is worth naming the coincidence.

As the object of study: an r -periodic function makes some unitary act as a single r -cycle on an

invariant subspace. Because $U^r = \mathbb{I}$ there, its eigenvalues are forced onto the grid $\{e^{2\pi i k/r}\}$ — the period becomes an eigenphase. This is §7.

As the tool: because the shift's eigenvectors *are* the Fourier basis, estimating its eigenphases is the same as applying $F_{\mathbb{Z}_N}$. This is the construction above.

The two statements are one theorem read in opposite directions: diagonalising the cyclic shift is the discrete Fourier transform.

6 The hidden subgroup problem

Definition. Given $f : G \rightarrow S$ on a known group G with

$$f(x) = f(y) \iff x^{-1}y \in H$$

for an unknown subgroup $H \leq G$, find H . We say f *hides* H .

Equivalently, f is constant on left cosets of H and distinct on distinct cosets. Both halves matter: constancy is what lets branches interfere, distinctness is what keeps different cosets from mixing. For abelian G the left/right distinction is vacuous.

Why this is classically hard

Theorem 6 (Childs, Thm. 5.1). *If G has a family $\mathcal{H}$ of N subgroups whose only common element is the identity, then a deterministic classical algorithm needs $\Omega(\sqrt{N})$ queries to solve the HSP.*

Proof sketch. Let the oracle behave adversarially: on the ℓ -th distinct query g_ℓ it answers ℓ whenever some $H \in \mathcal{H}$ remains consistent with all answers so far, and concedes otherwise. An algorithm making t queries sees the fixed response sequence $1, \dots, t$, so it cannot adapt. Forcing a concession requires that for every $H \in \mathcal{H}$ some pair satisfies $g_k g_j^{-1} \in H$; there are at most $t(t-1)$ such non-identity ratios against N subgroups, so $t(t-1) \geq N$, i.e. $t = \Omega(\sqrt{N})$. $\square$

The intuition is the birthday bound: with S unstructured, you learn nothing until you see a collision, and collisions need $\Omega(\sqrt{|G|/|H|})$ random queries. Note the converse — a group with only polynomially many subgroups (e.g. $\mathbb{Z}_p, \mathbb{Z}_{2^n}$) is classically *easy*, since one can simply test each candidate. Hardness comes from the number of subgroups, not from the group being large.

The standard recipe, and what it samples

- (1) Prepare the uniform superposition over G and query f : $|G|^{-1/2} \sum_x |x\rangle |f(x)\rangle$.
- (2) Discard the second register. Since f is injective across cosets, the states $|f(x)\rangle$ are orthogonal, so the first register is left in the uniform mixture over *coset states* $|cH\rangle$ with c uniformly random and unknown.
- (3) Apply F_G and measure.

The output is supported on the **annihilator**

$$H^\perp := \{y \in \hat{G} : \omega_y(h) = 1 \ \forall h \in H\}, \quad |H^\perp| = |G|/|H|, \quad (27)$$

and is *uniform* on it, because the unknown coset representative c contributes only an overall phase $\omega_y(c)$ and therefore nothing to $|\cdot|^2$. Since $H = (H^\perp)^\perp$, sampling enough of $H^\perp$ determines H .

Two consequences worth internalising. (a) Step (2) need not be performed. Measuring the second register produces exactly the ensemble that tracing it out produces; the orthogonal record forbids inter-coset interference either way. (b) The measured value is not “an approximate phase” but *a random element of a dual subgroup*. Approximation enters only when one is forced to transform over the wrong group — see §7 versus §9.

Part II — The two instances that break deployed cryptography

7 Order finding, and factoring

From factoring to a period

For N odd, composite and not a prime power, pick a with $\gcd(a, N) = 1$ and find the *order* $r = \text{ord}_N(a)$. If r is even and $x = a^{r/2} \not\equiv -1 \pmod{N}$, then minimality gives $x \not\equiv 1$, while $(x - 1)(x + 1) \equiv 0 \pmod{N}$; so $\gcd(x \pm 1, N)$ are proper factors. A random coprime a is favourable with probability at least $1/2$.

A sharper form of the “lucky a ” bound. For odd N with m distinct prime factors, the set of units a whose order r is even *and* satisfies $a^{r/2} \not\equiv -1 \pmod{N}$ has at least

$$\varphi(N) \left(1 - \frac{1}{2^{m-1}}\right)$$

elements; the proof rests on $(\mathbb{Z}/p^k\mathbb{Z})^\times$ being cyclic for odd primes p , plus the Chinese Remainder Theorem. Even numbers and prime powers are stripped off classically beforehand, so $m \geq 2$ always, giving the usual “at least half”. *The bound improves with the number of prime factors*, which is the opposite of the classical difficulty ordering — and it is why the hardest case for the classical stage, $N = pq$ a product of two large primes, is exactly the boundary case $m = 2$. (Verified exhaustively for $N = 15, 21, 33, 35, 105, 1155$; tight at $N = 21$ and $N = 33$.)

Periodicity is a cyclic orbit

On $n = \lceil \log_2 N \rceil$ qubits define modular multiplication

$$U_a|y\rangle = |ay \bmod N\rangle \quad (y < N), \quad U_a|y\rangle = |y\rangle \quad (N \leq y < 2^n), \quad (28)$$

a permutation matrix, hence unitary, with $U_a^r = \mathbb{I}$. On the orbit $K_1 = \text{span}\{|a^j\rangle : j = 0, \dots, r-1\}$ — whose r states are distinct, else minimality of r fails — U_a is a *single r -cycle*, i.e. the matrix

$$S_r = \begin{pmatrix} 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}, \quad (S_r)_{i,j} = \delta_{i,j+1 \bmod r}. \quad (29)$$

Lemma 7 (Eigenphases). $|\psi_k\rangle = \frac{1}{\sqrt{r}} \sum_{j=0}^{r-1} e^{-2\pi i j k / r} |a^j \bmod N\rangle$ are orthonormal with $U_a|\psi_k\rangle =$

$e^{2\pi i k/r} |\psi_k\rangle$, and

$$|1\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} |\psi_k\rangle. \quad (30)$$

Proof. The eigenrelation is the shift $j \mapsto j+1$ extracting $e^{2\pi i k/r}$; orthonormality is Lemma 2. For (30), exchange the finite sums: $\frac{1}{\sqrt{r}} \sum_k |\psi_k\rangle = \frac{1}{r} \sum_j (\sum_k e^{-2\pi i j k/r}) |a^j\rangle = \sum_j \delta_{j,0} |a^j\rangle = |1\rangle$, again by Lemma 2. $\square$

Two facts single out U_a . Its large powers are *cheap*: $U_a^{2^m} = U_{a^{2^m \bmod N}}$, with the constants precomputed classically by repeated squaring, so each controlled power is one modular multiplication. And its cyclic sector is *reachable*: preparing $|1\rangle$ costs one X gate, whereas preparing any individual $|\psi_k\rangle$ would require knowing r — the very unknown. Equation (30) says one never needs to.

Representation-theoretic reading. The orbit carries the *regular representation* of $\mathbb{Z}_r$, whose decomposition is multiplicity-free: every character occurs, and each occurs exactly once. That is the sharp form of “the single r -cycle makes every $e^{2\pi i k/r}$ occur.” Multiplicity one makes each eigenvalue non-degenerate, so the $|\psi_k\rangle$ are determined up to phase and the branch labels are canonical. Furthermore $|\psi_k\rangle$ is the isotypic projection $P_k|1\rangle$ with $P_k = \frac{1}{r} \sum_m e^{-2\pi i k m/r} U_a^m$, and $\sum_k P_k = \mathbb{I}$ applied to $|1\rangle$ is exactly (30). In Fourier language: $|1\rangle$ is a *delta function on the group*, and a *delta has flat spectrum* — which is why the eigenphase is sampled *exactly* uniformly below.

A dimension-count proof, when $r \mid M$. If $f : \mathbb{Z}_M \rightarrow \mathbb{C}$ has period r with $r \mid M$, then $\hat{f}(c) = 0$ except for $c \in \{0, M/r, 2M/r, \dots, (r-1)M/r\}$. *Proof:* the space of r -periodic functions on $\mathbb{Z}_M$ has dimension r , and the r characters $\chi_{M/r}, \chi_{2M/r}, \dots, \chi_{(r-1)M/r}, \chi_M = \mathbf{1}$ are r independent members of it, hence a basis. A function in the span of those characters has Fourier support inside their labels. $\square$

No computation at all: *periodicity is a subspace condition, and the characters that span that subspace are exactly the allowed frequencies.* The geometric-series calculation elsewhere in these notes is the same statement done by hand; this is the reason behind it.

The circuit, read in two bases

With a t -qubit control register, $Q = 2^t$, starting from $|0^t\rangle|1\rangle$ and applying Hadamards then controlled powers:

$$|Y\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle \otimes |a^x \bmod N\rangle \quad (\text{computational reading}) \quad (31)$$

$$= \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} |\Phi_{k/r}\rangle \otimes |\psi_k\rangle, \quad (\text{spectral reading}) \quad (32)$$

where $|\Phi_\theta\rangle = Q^{-1/2} \sum_x e^{2\pi i x \theta} |x\rangle$. The second line follows by inserting (30), applying U_a^x branch-wise and swapping sums. One state, two descriptions: modular exponentiation evaluated in superposition, or r phase ramps each tagged by an orthogonal eigenlabel.

The measurement law

Applying $F_Q^\dagger \otimes \mathbb{I}$ to (32) and expanding with (11):

$$|Y'\rangle = \sum_{y=0}^{Q-1} |y\rangle \otimes \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} A_y(k/r) |\psi_k\rangle. \quad (33)$$

Let K denote the outcome that *would* be obtained by measuring the work register in the eigenbasis. This is a legitimate random variable even though it is never measured: $|y\rangle\langle y| \otimes \mathbb{I}$ and $\mathbb{I} \otimes |\psi_k\rangle\langle\psi_k|$ act on different tensor factors, hence commute and are jointly measurable; and no operation on the work register can change the statistics of y . Since the coefficients live in an orthonormal product basis, $|\frac{1}{\sqrt{r}}A_y(k/r)|^2$ is a joint distribution, and

$$\Pr[K = k] = \frac{1}{r} \quad \text{exactly,} \quad \Pr[Y = y \mid K = k] = |A_y(k/r)|^2, \quad (34)$$

$$\Pr[Y = y] = \frac{1}{r} \sum_k |A_y(k/r)|^2, \quad \Pr[K = k \mid Y = y] = \frac{|A_y(k/r)|^2}{\sum_h |A_y(h/r)|^2}. \quad (35)$$

The first marginal uses $\sum_y |A_y(\theta)|^2 = 1$ for *every* real θ (the transform of a unit vector is a unit vector) — which also makes the conditional a genuine normalised distribution, so that *conditioned on the branch, the control register sees textbook phase estimation on a true eigenstate*, with no correction for having started at $|1\rangle$. The third is the only place orthogonality does real work:

$$\Pr[Y = y] = \langle Y' | (|y\rangle\langle y| \otimes \mathbb{I}) | Y' \rangle = \frac{1}{r} \sum_{k,k'} \overline{A_y(k'/r)} A_y(k/r) \underbrace{\langle \psi_{k'} | \psi_k \rangle}_{\delta_{kk'}} = \frac{1}{r} \sum_k |A_y(k/r)|^2, \quad (36)$$

the inter-branch cross terms being annihilated by orthogonality and nothing else.

Operationally: the statistics are exactly those of *draw k uniformly, then run ideal phase estimation on k/r* . This is an equality of statistics, not a hidden classical choice — the circuit stays coherent throughout, but the work register is a perfect (unread) which-branch record, so the r ramps coexist without blurring and eigenphases are *sampled*, never averaged.

The general case: ragged fibres and the exact amplitude

The clean statements above assume $r \mid Q$. Generically it does not, and the honest bookkeeping is worth seeing once because it is where the approximation genuinely enters.

Write $Q = qr + b$ with $0 \leq b < r$. The fibre of $f(x) = a^x \bmod N$ over a value a^s is the arithmetic progression

$$f^{-1}(a^s) = \{s + kr : k = 0, \dots, K_s - 1\}, \quad K_s = \begin{cases} q + 1, & s < b, \\ q, & s \geq b, \end{cases} \quad (37)$$

so the r fibres have *unequal* sizes differing by one. (Verified: at $N = 21$, $a = 11$, $r = 6$, $Q = 2^{10}$ one gets $Q = 170 \cdot 6 + 4$ and fibre sizes 171, 171, 171, 171, 170, 170.) Measuring the work register — or equivalently, by §6, not measuring it — leaves the first register in the normalised indicator $\psi_s = \mathbf{1}_{f^{-1}(a^s)} / \sqrt{K_s}$, and the exact outcome law is

$$|\hat{\psi}_s(c)|^2 = \frac{1}{K_s Q} \left| \sum_{k=0}^{K_s-1} \exp\left(-\frac{2\pi i c (s + kr)}{Q}\right) \right|^2. \quad (38)$$

Two features deserve notice. The coset representative s enters only through the factor $e^{-2\pi i c s / Q}$ of modulus one, so it *cannot* affect the probabilities — the same mechanism as in §9, here in its number-theoretic guise. And the geometric series is truncated at K_s rather than closing on a full period, which is exactly why the peaks acquire width: this is *spectral leakage*, the discrete-Fourier phenomenon of analysing a signal over a window that is not a whole number of periods.

Numerically, at $N = 15$, $a = 7$ ($r = 4 \mid Q$) the outcome distribution is supported on exactly 4 values with total mass 1; at $N = 21$, $a = 11$ ($r = 6 \nmid Q$) it is supported on *all* Q values, with 0.789 of the mass on the r nearest grid points — and that fraction is the same to four decimals for $Q = 2^7$ and $Q = 2^{10}$. Enlarging the register sharpens the resolution but does not change the fraction of mass in the main lobes; that constant is what Lemma 3 bounds below by $4/\pi^2$.

The combined bound. Packaging the Dirichlet estimate with the coprimality condition, the probability of observing c for which there exists s with $0 \leq s < r$, $\gcd(s, r) = 1$ and

$$\left| \frac{c}{Q} - \frac{s}{r} \right| < \frac{1}{2r^2}$$

is at least

$$\frac{4}{\pi^2} \cdot \frac{\varphi(r)}{r} \cdot \left(1 - \frac{1}{N}\right).$$

The three factors are the three separate things that must go right: the Fourier peak must be resolved ($4/\pi^2$), the sampled numerator must be coprime to r ($\varphi(r)/r$), and the window must be long enough ($1 - 1/N$).

How small can $\varphi(r)/r$ get?

The $\varphi(r)/r$ factor is the only one that degrades with N , so it deserves a precise statement rather than the usual hand-wave. Both directions are known:

$$\frac{\varphi(m)}{m} < \frac{1}{e^\gamma \log \log m} \text{ for infinitely many } m, \quad \frac{\varphi(m)}{m} > \frac{1}{e^\gamma \log \log m + \frac{2.50637}{\log \log m}} \quad (m > 2), \quad (39)$$

with $\gamma = 0.5772 \dots$ the Euler–Mascheroni constant; the lower bound is Rosser–Schoenfeld. So the decay is real — it is not an artefact of a lossy analysis — but it is $\Theta(1/\log \log m)$, which is savage in asymptotic notation and negligible in practice: the bound is still 0.135 at $m = 2^{64}$. A constant number of repetitions suffices at any size one could actually run.

Classical closing

Take $t = 2n$, so $Q \geq N^2$. By Lemma 3, with probability $\geq 4/\pi^2$,

$$\left| \frac{y}{Q} - \frac{k}{r} \right| \leq \frac{1}{2Q} \leq \frac{1}{2N^2} < \frac{1}{2r^2}. \quad (40)$$

Legendre’s criterion — any p/q with $|x - p/q| < 1/(2q^2)$ is a continued-fraction convergent of x — then puts k/r in lowest terms among the $O(n)$ convergents of y/Q . The recovered denominator is $r/\gcd(k, r)$, so a run succeeds outright when $\gcd(k, r) = 1$, with probability $\varphi(r)/r = \Omega(1/\log \log N)$; or take the lcm of two runs’ denominators, which is r with probability $\prod_{p|r} (1 - p^{-2}) \geq 6/\pi^2$.

8 The algorithm in full: $N = 15, a = 7$

Everything above is abstract. The smallest instance that exhibits the whole mechanism is $N = 15$ with $a = 7$; because its period is a power of two, every approximation collapses and the algebra can be written out exactly. All states below were computed directly.

Setup

Take $n = \lceil \log_2 15 \rceil = 4$ work qubits and $t = 2n = 8$ control qubits, so $Q = 2^8 = 256$. The function $f(x) = 7^x \bmod 15$ cycles through

$$1, 7, 4, 13, 1, 7, 4, 13, \dots \implies r = 4, \quad (41)$$

so $r \mid Q$ and each of the 4 fibres has exactly $L = Q/r = 64$ elements.

The permutation, and why only four eigenvectors matter

Multiplication by 7 modulo 15 permutes $\{0, \dots, 15\}$ in disjoint cycles:

$$(1\ 7\ 4\ 13), \quad (2\ 14\ 8\ 11), \quad (3\ 6\ 12\ 9), \quad \text{fixed points } 0, 5, 10, 15. \quad (42)$$

U_a is a 16×16 permutation matrix and therefore has 16 eigenvectors — yet only four ever appear. The reason is that $|1\rangle$ lies in the 4-dimensional invariant subspace spanned by its own cycle, and the other cycles span subspaces with *zero overlap*. The circuit never leaves the sector it starts in.

A consequence worth noting. Any cycle would do: seeding the work register with $|2\rangle$ would run the identical algorithm inside $\{2, 14, 8, 11\}$ and return the same r . In practice 1 is the only element of the orbit one can name without already knowing something about the group, which is why $|1\rangle$ is the universal choice — not because it is special, but because it is *identifiable*. (Note also the shorter cycles: 5 and 10 are fixed because $\text{ord}_{15/5}(7) = 1$; a general orbit of z has length $\text{ord}_{N/\text{gcd}(z,N)}(a)$, which divides r .)

The four relevant eigenvectors are $|u_z\rangle = \frac{1}{2} \sum_{j=0}^3 i^{-jz} |7^j\rangle$:

	in the computational basis	eigenvalue	phase $\theta_z = z/4$
$ u_0\rangle$	$\frac{1}{2}(1\rangle + 7\rangle + 4\rangle + 13\rangle)$	+1	0
$ u_1\rangle$	$\frac{1}{2}(1\rangle - i 7\rangle - 4\rangle + i 13\rangle)$	+i	1/4
$ u_2\rangle$	$\frac{1}{2}(1\rangle - 7\rangle + 4\rangle - 13\rangle)$	-1	1/2
$ u_3\rangle$	$\frac{1}{2}(1\rangle + i 7\rangle - 4\rangle - i 13\rangle)$	-i	3/4

and, as Lemma 7 promises, $\frac{1}{2}(|u_0\rangle + |u_1\rangle + |u_2\rangle + |u_3\rangle) = |1\rangle$ (checked exactly). This is Case 2 of §4 with four branches.

The state, step by step

After the Hadamards and the oracle, grouping the 256 terms by the value in the work register:

$$\begin{aligned} |\Psi_2\rangle = \frac{1}{16} & \left[\underbrace{(|0\rangle + |4\rangle + |8\rangle + \dots + |252\rangle)}_{64 \text{ terms}} |1\rangle + (|1\rangle + |5\rangle + \dots + |253\rangle) |7\rangle \right. \\ & \left. + (|2\rangle + |6\rangle + \dots + |254\rangle) |4\rangle + (|3\rangle + |7\rangle + \dots + |255\rangle) |13\rangle \right]. \end{aligned} \quad (43)$$

Each work value is entangled with an arithmetic progression of step $r = 4$ in the control register. The period is present, but unreadable: measuring now gives a uniformly random $x \in \{0, \dots, 255\}$.

The inverse transform converts each progression of *step 4* into a comb of *four* peaks at spacing $Q/r = 64$:

$$|\Psi_3\rangle = \frac{1}{4} \left[(|0\rangle + |64\rangle + |128\rangle + |192\rangle) |1\rangle + (|0\rangle - i|64\rangle - |128\rangle + i|192\rangle) |7\rangle + \dots \right] \quad (44)$$

and the entire state is captured by a 4×4 table of coefficients (all equal to $\frac{1}{4}$ times a fourth root of unity):

$4 \langle k, d \Psi_3 \rangle$	$d = 1$	$d = 7$	$d = 4$	$d = 13$
$k = 0$	1	1	1	1
$k = 64$	1	-i	-1	+i
$k = 128$	1	-1	1	-1
$k = 192$	1	+i	-1	-i

The table is the Fourier matrix — and it is the whole algorithm

Read that array twice.

By columns (fix a work value d): each column is a *phase ramp* over the four peaks — the kicked-back eigenphase, exactly (20).

By rows (fix a peak k): each row is an *eigenvector*. Row $k = 0$ is $|u_0\rangle$, row $k = 64$ is $|u_1\rangle$, row $k = 128$ is $|u_2\rangle$, row $k = 192$ is $|u_3\rangle$.

The array with entries ω_r^{-jz} is precisely the $r \times r$ discrete Fourier matrix, so

$$|\Psi_3\rangle = \frac{1}{\sqrt{r}} \sum_{z=0}^{r-1} \left| z \frac{Q}{r} \right\rangle \otimes |u_z\rangle \quad (45)$$

is a Schmidt decomposition of rank r with equal coefficients. **The entanglement between the two registers is exactly the Fourier duality between position (which remainder) and frequency (which peak)**, and the correlation $|zQ/r\rangle \leftrightarrow |u_z\rangle$ is perfect. This is the abstract statement of §7 made completely explicit — and it shows at a glance why measuring the work register is optional: whichever remainder it collapses to, the control register is left in a superposition of the same four peaks.

Reading it out

Each of $k \in \{0, 64, 128, 192\}$ occurs with probability exactly $1/4$. Given k , one has $k/Q = z/r$: e.g. $k = 192$ gives $192/256 = 3/4$, whose reduced denominator is $r = 4$. Then $7^2 = 49 \equiv 4 \pmod{15}$ and

$$\gcd(4 - 1, 15) = 3, \quad \gcd(4 + 1, 15) = 5. \quad (46)$$

The failure mode is $k = 0$, carrying no information — here with probability $1/4$, since r is tiny. Note also that $k = 128$ gives $1/2$, whose denominator is 2: a proper divisor of r , i.e. the $\gcd(z, r) \neq 1$ case, which is why the classical stage must verify $a^{r'} \equiv 1$ and combine samples.

Why this instance is misleadingly easy. $r = 4$ is a power of two, so $r \mid Q$ and the peaks are exact — no leakage, no continued fractions, and t could be reduced to $\lceil \log_2 r \rceil = 2$ qubits. The contrast with $N = 21$, $a = 11$ ($r = 6 \nmid Q$) is stark: there the outcome distribution has support on *all* Q values, with only ≈ 0.79 of the mass in the main lobes, and continued fractions become necessary. Demonstrations built on $N = 15$ therefore hide precisely the difficulty that §7 exists to handle.

9 Discrete logarithms

The problem, and why it is at least as hard as factoring

Let $G = \langle g \rangle$ be cyclic with *known* order $N = |G|$, and let $x \in G$. The discrete logarithm $\log_g x$ is the least $\lambda \geq 0$ with $g^\lambda = x$. Computing g^λ is easy by repeated squaring; inverting it is not known to be classically feasible. Diffie–Hellman key exchange rests on exactly this asymmetry.

Factoring reduces to discrete log in $\mathbb{Z}_N^\times$ — though not usefully as a quantum algorithm, since the reduction needs $|\mathbb{Z}_N^\times| = \varphi(N)$, whose computation is as hard as factoring. So discrete log is the harder-looking of the two, and yet it has the *simpler* quantum algorithm.

As a hidden subgroup problem

Take $G_{\text{HSP}} = \mathbb{Z}_N \times \mathbb{Z}_N$ and

$$f(\alpha, \beta) = x^\alpha g^\beta = g^{\alpha \log_g x + \beta}. \quad (47)$$

Then f depends only on $\alpha \log_g x + \beta$, so it is constant on the lines $L_\gamma = \{(\alpha, \beta) : \alpha \log_g x + \beta = \gamma\}$ and hides

$$H = L_0 = \langle (1, -\log_g x) \rangle = \{(\alpha, -\alpha \log_g x) : \alpha \in \mathbb{Z}_N\}, \quad |H| = N. \quad (48)$$

Knowing H is knowing $\log_g x$: it is minus the second coordinate of the generator whose first coordinate is 1. *Geometrically the hidden subgroup is a line in a discrete torus and the secret is its slope* — which is why this instance needs a rank-two group, whereas a period is a rank-one phenomenon.

The derivation

Start from the uniform superposition and query:

$$\frac{1}{N} \sum_{\alpha, \beta \in \mathbb{Z}_N} |\alpha, \beta\rangle \mapsto \frac{1}{N} \sum_{\alpha, \beta} |\alpha, \beta, f(\alpha, \beta)\rangle. \quad (49)$$

Discard the third register. Conditioned on value g^γ the first two hold the coset state

$$|L_\gamma\rangle = \frac{1}{\sqrt{N}} \sum_{\alpha \in \mathbb{Z}_N} |\alpha, \gamma - \alpha \log_g x\rangle. \quad (50)$$

The observed value is useless on its own — reading γ off g^γ is the very problem we are solving — which is exactly why discarding costs nothing. Now apply $F_{\mathbb{Z}_N} \otimes F_{\mathbb{Z}_N}$ (one transform over $\mathbb{Z}_N \times \mathbb{Z}_N$, by Lemma 1) and use Lemma 2:

$$\frac{1}{N^{3/2}} \sum_{\alpha, \mu, \tau} \epsilon_N^{\mu\alpha + \tau(\gamma - \alpha \log_g x)} |\mu, \tau\rangle = \frac{1}{N^{3/2}} \sum_{\mu, \tau} \epsilon_N^{\tau\gamma} \underbrace{\left(\sum_{\alpha} \epsilon_N^{\alpha(\mu - \tau \log_g x)} \right)}_{= N [\mu \equiv \tau \log_g x]} |\mu, \tau\rangle \quad (51)$$

$$= \frac{1}{\sqrt{N}} \sum_{\tau \in \mathbb{Z}_N} \epsilon_N^{\tau\gamma} |\tau \log_g x, \tau\rangle \quad (52)$$

The unknown γ survives *only* as a phase and therefore not at all in the statistics. Measuring gives $(\tau \log_g x, \tau)$ for uniformly random $\tau \in \mathbb{Z}_N$; the support is precisely $H^\perp = \langle (\log_g x, 1) \rangle$, as §6 predicts. If τ is invertible, divide:

$$\log_g x = \mu \tau^{-1} \bmod N, \quad (53)$$

which happens with probability $\varphi(N)/N = \Omega(1/\log \log N)$; otherwise repeat. (Verified numerically at $N = 7$ and $N = 12$: N outcomes, uniform, all on the dual line, with invertible- τ fractions $6/7$ and $4/12 = \varphi(12)/12$ respectively.)

The elliptic-curve specialisation

Nothing above used more than: G cyclic of *known* order, with efficiently computable products. Writing the group additively with base point P of order r and public point $Q = [m]P$, the same algorithm runs with $f(k, \ell) = [k]P + [\ell]Q = [k + m\ell]P$, hidden subgroup $L = \langle (-m, 1) \rangle$, and output uniform on $L^\perp = \langle (1, m) \rangle$. So ECC and finite-field Diffie–Hellman fall to *one* theorem, not two coincidences.

	multiplicative (Childs, §9)	additive / ECC
generator, public element	$g, \quad x = g^{\log_g x}$	$P, \quad Q = [m]P$
secret	$\log_g x$	m
indices	(α, β)	(ℓ, k)
hiding function	$x^\alpha g^\beta$	$[k]P + [\ell]Q$
hidden subgroup	$\langle (1, -\log_g x) \rangle$	$\langle (-m, 1) \rangle$
annihilator	$\langle (\log_g x, 1) \rangle$	$\langle (1, m) \rangle$
recovery	$\mu \tau^{-1}$	$y_2 y_1^{-1}$

The two differ only by the ordering of the two index registers; under $(\alpha, \beta) \leftrightarrow (\ell, k)$ every row matches.

Where the two instances genuinely differ

Both algorithms must sample a *generator* of a cyclic group of order r , and this is the single axis along which they diverge.

Order finding: r is the unknown. One must therefore transform over $\mathbb{Z}_{2^i}$ rather than $\mathbb{Z}_r$, so Lemma 2 applies only approximately — hence the Dirichlet kernel, the $4/\pi^2$ bound and continued fractions. And r is composite, so a uniform sample generates only with probability $\varphi(r)/r$: repetition, lcm-combining, gcd conditions on a .

Discrete log: N (equivalently r) is *public*. One transforms over the actual group; Lemma 2 is an exact delta and the post-processing is one modular inversion. For general N the generator lottery remains, at $\varphi(N)/N$; but deployed elliptic curves use *prime-order* subgroups, where every non-identity element generates, so the idealised success probability is $1 - 1/r$.

Caveat, and a correction. A real implementation still uses a power-of-two transform, because the staircase circuit of §2 is what one can build cheaply. That reintroduces the Dirichlet blur — but not continued fractions, since knowing r permits *rescale-and-round*: $u_i = \text{round}(y_i r / 2^w) \bmod r$ with $w = \lceil \log_2 r \rceil + 2 \dots 4$. Published circuit designs report 0.7–0.8 success per shot, so roughly ten shots. The exact single-shot result is the idealised $F_{\mathbb{Z}_r}$ statement; it should not be quoted as an implementation figure.

Part III — The interference audit

10 Records, and the master formula

Everything so far assumed the oracle leaves nothing behind but its output. Real circuits generate enormous quantities of intermediate data, and the question of which of it is fatal has an exact answer.

Take the discrete-log setting on $\mathbb{Z}_r \times \mathbb{Z}_r$ and carry an extra scratch register through:

$$|\Psi\rangle = \frac{1}{r} \sum_{x_1, x_2 \in \mathbb{Z}_r} |x_1\rangle |x_2\rangle |f(x)\rangle_{\text{acc}} |g(x)\rangle_{\text{scr}}, \quad f(x) = [x_1 + kx_2]P. \quad (54)$$

Apply $F_r \otimes F_r$ and group by the value of the index registers:

$$\sum_{y_1, y_2} |y_1\rangle |y_2\rangle \otimes \underbrace{\frac{1}{r^2} \sum_x \epsilon_r^{x_1 y_1 + x_2 y_2} |f(x)\rangle |g(x)\rangle}_{|A_y\rangle}. \quad (55)$$

Because only the index registers are measured, the outcome probability is the *squared norm* of the leftover vector, $\Pr[y] = \langle A_y | A_y \rangle$ — not the modulus-squared of a scalar. Expanding $\|\sum_x v_x\|^2 = \sum_{x,x'} \langle v_{x'} | v_x \rangle$ gives

Master formula: one damping factor per unmeasured register

$$\Pr[y_1, y_2] = \frac{1}{r^4} \sum_{x, x'} \varepsilon_r^{(x_1 - x'_1)y_1 + (x_2 - x'_2)y_2} \underbrace{\langle f(x') | f(x) \rangle}_{= [c(x)=c(x')]} \underbrace{\langle g(x') | g(x) \rangle}_{\text{should be } \equiv 1} \quad (56)$$

The double sum ranges over *pairs* of computational paths: $x = x'$ is the classical part, $x \neq x'$ the interference. The phase depends only on the difference $x - x'$ and is what can cancel. Each unmeasured register contributes one overlap factor multiplying the cross terms — and both records enter *in the same slot, in the same way*.

The accumulator holds a computational basis state, so its overlap is a Kronecker delta on the point value, which by injectivity across cosets equals $[c(x) = c(x')]$. Restricting to same-coset pairs and parametrising a coset by $\ell := x_2$ (so $x_1 = c - k\ell$), the phase becomes $\varepsilon_r^{cy_1} \varepsilon_r^{\ell(y_2 - ky_1)}$ and the c -dependence cancels between x and x' , leaving with $d := y_2 - ky_1$

$$\Pr[y] = \frac{1}{r^4} \sum_{c \in \mathbb{Z}_r} \sum_{\ell, \ell' \in \mathbb{Z}_r} \varepsilon_r^{(\ell - \ell')d} \langle g_{c, \ell'} | g_{c, \ell} \rangle. \quad (57)$$

11 The damage law

No record

If every branch leaves the scratch in the *same* state (in practice $|0 \cdots 0\rangle$), then $\langle g_{\ell'} | g_{\ell} \rangle = \langle g | g \rangle = 1$ for *every* pair, including the off-diagonal ones. The constant pulls out and the double sum **factorises**:

$$\sum_{\ell, \ell'} \varepsilon_r^{(\ell - \ell')d} \cdot 1 = \left(\sum_{\ell} \varepsilon_r^{\ell d} \right) \overline{\left(\sum_{\ell'} \varepsilon_r^{\ell' d} \right)} = \left| \sum_{\ell} \varepsilon_r^{\ell d} \right|^2 = r^2 [d \equiv 0], \quad (58)$$

giving $\Pr[y] = \frac{1}{r} [y_2 \equiv ky_1]$. That factorisation is what interference is: amplitudes summed first, squared second. Note the content is not that $\langle g | g \rangle = 1$ — a normalised state always satisfies that — but that it is the *same* 1 on the off-diagonal, i.e. that the subscript is absent.

A record that partitions each coset

Suppose the scratch takes orthogonal values on blocks $B_1, \dots, B_t$ partitioning the r branches of a coset, $|B_{\beta}| = n_{\beta}$. Then $\langle g_{\ell'} | g_{\ell} \rangle$ is the indicator of “same block” and the double sum splits into one character sum per block, added *incoherently*:

$$\sum_{\ell, \ell'} \varepsilon_r^{(\ell - \ell')d} [\text{same block}] = \sum_{\beta=1}^t \left| \sum_{\ell \in B_{\beta}} \varepsilon_r^{\ell d} \right|^2. \quad (59)$$

This is the whole problem. Lemma 2 cancels the *complete* sum over $\mathbb{Z}_r$. A *partial* sum over a proper subset does not vanish. So for $d \not\equiv 0$ probability leaks off the dual line, and at $d \equiv 0$ each block contributes n_{β} rather than the blocks contributing r jointly.

Hence

$$\Pr[\text{peak}] = \frac{1}{r^3} \sum_{\beta} n_{\beta}^2, \quad \boxed{\frac{\text{signal on the dual line}}{\text{ideal}} = \frac{\sum_{\beta} n_{\beta}^2}{r^2}} \quad (60)$$

By Cauchy–Schwarz $\sum_{\beta} n_{\beta}^2 \leq (\sum_{\beta} n_{\beta})^2 = r^2$, with **equality iff** $t = 1$. Any record that splits a coset at all strictly reduces the signal, and the loss depends only on how the branches are partitioned. Two corollaries: a balanced two-way split gives $2(r/2)^2/r^2 = \frac{1}{2}$, so *one leaked bit halves the signal*; a full leak gives $r/r^2 = 1/r$, i.e. the uniform distribution over all r^2 pairs and zero information.

leaked	block sizes	peak (simulated)	peak, (60)	signal (sim.)	$\sum n_{\beta}^2/r^2$
nothing	[7]	0.14286	0.14286	1.00000	49/49
parity of x_1	[4, 3]	0.07289	0.07289	0.51020	25/49
$x_1 \bmod 3$	[3, 2, 2]	0.04956	0.04956	0.34694	17/49
x_1 entirely	[1 ⁷]	0.02041	0.02041	0.14286	7/49

Direct simulation at $r = 7, k = 4$ against the closed form; agreement is exact.

Partially distinguishable records

If the records overlap rather than being orthogonal, $\langle g_{\ell'} | g_{\ell} \rangle = \delta_{\ell\ell'} + s(1 - \delta_{\ell\ell'})$, then

$$\sum_{\ell, \ell'} \epsilon_r^{(\ell - \ell')d} [\delta_{\ell\ell'} + s(1 - \delta_{\ell\ell'})] = (1 - s)r + sr^2[d \equiv 0], \quad (61)$$

$$\Pr[y] = \underbrace{\frac{s}{r} [y_2 \equiv ky_1]}_{\text{ideal}} + \underbrace{\frac{1-s}{r^2}}_{\text{uniform}}. \quad (62)$$

The output is *exactly* the ideal distribution and white noise mixed with weights s and $1 - s$. The same law holds for phase estimation, where it reads $\Pr[y] = s |A_y(\theta)|^2 + (1 - s)/Q$ — verified numerically to machine precision. Useful signal = $s + (1 - s)/r \approx s$: linear in the record overlap, with no threshold to hide behind.

12 The two-sided rule

Formula (56) treats both unmeasured registers identically, which means the design task is to arrange *opposite* values for them:

$$\langle f(x') | f(x) \rangle = \delta_{c,c'} \quad \text{a record you must *protect*,} \quad (63)$$

$$\langle g(x') | g(x) \rangle \equiv 1 \quad \text{a record you must *destroy*.} \quad (64)$$

Clear the scratch; keep the accumulator

Clear every record that distinguishes branches you want to interfere. The r same-coset branches must remain indistinguishable, or the character sum never fires.

*Keep the record that distinguishes branches you must **not** let interfere.* The accumulator’s orthogonality is what stops different cosets from blurring together — and it is why measuring it is optional.

A register may survive to the transform **iff its contents are a function of the coset label alone**. The accumulator qualifies; nothing else does. Apply “always uncompute” uniformly and you restore $|P_0\rangle$ on every branch, making the state a product, whose inverse transform is $|0\rangle$: you measure $(0, 0)$ with certainty. Apply “records are harmless if unread” and you get uniform noise. Nothing in the syntax of a reversible circuit marks which register is which; it is a property of the *algorithm*.

Why scratch varies within a coset

It is tempting to think that branches converging on the same point compute “morally the same thing”. They do not. With $r = 7$, $k = 4$, the branches $(x_1, x_2) = (0, 0)$ and $(3, 1)$ lie in the same coset, yet:

step	branch (0, 0)	branch (3, 1)
seed	P_0	P_0
$+ [1]P$ (bit $x_1[0]$)	—	$P_0 + P$
$+ [2]P$ (bit $x_1[1]$)	—	$P_0 + 3P$
$+ [1]Q$ (bit $x_2[0]$)	—	$P_0 + 3P + 4P = P_0$
final	P_0	P_0

They agree at the start and at the end and *nowhere in between*. Every intermediate accumulator value differs, so every slope differs, so every modular inversion sees a different input, so every branch history, product and comparison flag differs. Convergence is a property of the final value only — which is why the volume of garbage is enormous and every bit of it separates branches required to add coherently.

The timing matters and is easy to misstate: the rule is *not* “never entangle”. Intermediate entanglement is unavoidable and harmless. The requirement is only that at the instant the transform runs, all scratch is back at $|0\rangle$.

Why “clear” can only mean uncompute

Three plausible alternatives fail, each instructively:

- **Discard it.** Ignoring a register *is* tracing it out, and the overlap factor enters $\Pr[y]$ regardless of whether anyone looks. The table above is the discard case throughout.
- **Measure and reset.** Measurement collapses onto one value of g , projecting onto the branches consistent with it. That does not restore interference; it discards the rest of the superposition.
- **Overwrite with $|0\rangle$.** Not unitary on a register entangled with the rest of the state — a reset channel, i.e. decoherence with extra steps.

Only running the generating computation backwards works:

$$\sum_{\ell} \alpha_{\ell} |\ell\rangle |g_{\ell}\rangle \xrightarrow{U^{\dagger}} \left(\sum_{\ell} \alpha_{\ell} |\ell\rangle \right) \otimes |0\rangle, \quad (65)$$

a single unitary returning the ancilla to $|0\rangle$ on *every branch simultaneously*, leaving all amplitudes and relative phases intact. This is Bennett’s compute–copy–uncompute, and in circuit terms the *conjugation block* $U^{\dagger} V U$.

Part IV — Realisation and consequences

13 What a real ECDLP circuit looks like

Let $E : y^2 = x^3 + ax + b$ over $\mathbb{F}_p$, $n = \text{bitlen}(p)$, base point G of prime order r , public point $P = kG$. Note the two distinct sizes: n measures the *field* and sizes all arithmetic; the exponent registers have width $w = \lceil \log_2 r \rceil + 2 \dots 4$, set by the *group order*. Hasse’s theorem gives $r \approx p$ on a prime-order curve, so $w \approx n$.

The top level

Four registers: two exponent registers $x_1[w], x_2[w]$ and the two affine coordinates $AX[n], AY[n]$ of one accumulator point.

- (1) **Seed** X gates load $P_0 = c_0G$ for random $c_0 \in [1, r - 1]$. Free.
- (2) $H^{\otimes w}$ on both exponent registers.
- (3) **Precompute** (off-circuit): the host builds the doubling chains $[2^i]G$ and $[2^i]P$. Costs $w - 1$ classical field inversions, *zero qubits, zero gates*.
- (4) **Two ladders**: w controlled additions of $[2^i]G$ by the bits of x_1 , then w of $[2^i]P$ by the bits of x_2 , giving $|P_0\rangle \mapsto |(c_0 + x_1 + kx_2 \bmod r)G\rangle$.
- (5) **Inverse QFT** on each exponent register.
- (6) **Measure** both. The accumulator is *discarded*.

Three structural decisions

(a) The base point is classical. Since $x_1G = \sum_i (x_1)_i [2^i]G$ and the constants do not depend on x_1 , they are precomputed on the host. Hence no point is ever *doubled* on the quantum computer and no addition has two quantum summands: every one of the $2w$ additions is (quantum accumulator) + (classical constant), gated by one qubit.

Price: only the generic chord formula $\lambda = (y_2 - y_1)/(x_2 - x_1)$ is implemented, which breaks when the addend shares an x -coordinate with the accumulator. Handling that needs a data-dependent branch (expensive reversibly) or the complete addition formulas (far costlier), so it is simply not handled: a run fails with probability $O(n/r)$ by a union bound over $2w$ additions. At $n = 256$, $r \approx 2^{256}$, that is $\sim 10^{-75}$; at toy sizes it is *dominant*, which is why small-instance demonstrations must use an idealised oracle rather than this circuit. The random seed exists for the same reason: $(0, 0)$ is not a valid point encoding, and a *small* c_0 makes early additions exceptional on half the branches (seed $c_0 = 1$ and the first controlled $+ [2^0]G$ is a doubling).

(b) Addition is in place. Each addition's output feeds the next. Out of place costs either $O(n^2)$ qubits or double the gates to uncompute inputs; in place pins the accumulator at $2n$.

(c) The accumulator is never uncomputed — §12 made operational. Uncomputing it would restore $|P_0\rangle$ on every branch and return $(0, 0)$ with certainty. Discarding also halves the circuit.

The layer stack

layer	routine	cost	calls downward
L0	Shor driver	$\Theta(n^3)$	$\times 2$ scalar multiplications
L1	scalar multiplication	$\Theta(n^3)$	$\times n$ point additions (build-time unroll)
L2	point addition, in place	$\Theta(n^2)$	$\times 4$ inversions, $\times 20$ multiplications
L3a	modular inversion (Kaliski)	$\Theta(n^2)$	$2n$ rounds, $O(1)$ modular ops each
L3b	modular multiplication (Montgomery)	$\Theta(n^2)$	$\approx 2n$ ladder additions
L4	modular add/sub/negate/halve	$\Theta(n)$	$\times 5$ plain adders
L5	plain binary adder, in place	$\Theta(n)$	$\times (n - 1)$ AND gadgets
L6	temporary AND gadget	4 T	4 T compute, 0 T uncompute

L1 *emits no gates of its own*: it is a build-time unroll over the precomputed table, so the circuit is flat and the “recursion” describes how it was generated, not runtime behaviour. L3a runs a *fixed* $2n$ rounds because one cannot branch on data reversibly; each round's decision goes into a $2n$ -qubit *branch history*, the purest which-path record in the circuit. L4's five adders are “one add, one reduce, three to clear the flags”.

Reversibility is the cost driver

Roughly half the work at every layer is undoing, and four instances illustrate the whole pattern: modular addition spends three of five adders returning flags to $|0\rangle$; the inversion’s branch history clears only by running the inversion backwards, so the caller pays twice; point addition’s four inversion calls contribute *no bit* to the output, all sitting inside conjugation blocks; and the AND gadget uncomputes for zero T — by measuring in the X basis and applying a classically-conditioned Clifford correction.

That last asymmetry is load-bearing. In surface-code fault tolerance the Clifford gates are nearly free and the non-Clifford T gate dominates, because it needs magic-state distillation; so T-count is the currency. Without measurement-based uncomputation the deeply nested compute/uncompute structure above would roughly double the T-count. The reversibility tax is not an inefficiency to optimise away — *it is the invoice for the character sum of §10* — but the AND gadget moves it off the resource that matters.

Where n^3 comes from

Multiply along any single path. Exactly three nested factors of n :

$$\underbrace{2}_{\text{registers}} \times \underbrace{n}_{\text{point adds per scalar mult}} \times \underbrace{(4+20)}_{\text{L3 calls per point add}} \times \underbrace{\Theta(n)}_{\text{L4/L5 calls per L3 call}} \times \underbrace{5}_{\text{adders per mod op}} \times \underbrace{(n-1)}_{\text{ANDs per adder}} \times 4T = \Theta(n^3)T \quad (66)$$

namely one exponent bit per point addition, one field-size loop inside inversion/multiplication, and one carry per bit inside the adder. Everything else is a constant. The published figure is $\approx 5.8 \times 10^{10} T$ at $n = 256$.

Resources and the interface contract

$4n$ qubits persistent (two exponent registers, two accumulator coordinates); one addition live at a time adds a slope register, the $2n$ -bit history, a product scratch and the inversion’s own $3n$, so plan $10n$ – $11n$, i.e. ≈ 2800 logical qubits at $n = 256$ — consistent with the standard estimate of $9n + 2\lceil \log_2 n \rceil + 10 \approx 2330$ for n -bit ECDLP, against ≈ 6000 logical for RSA-2048.

The interface contract is what makes the layering sound and is worth copying into any implementation: *every boundary passes little-endian registers holding plain residues in $[0, p)$ — no Montgomery factor, no infinity flag — with all ancillas at $|0\rangle$ on entry and exit.* The last clause is §12 as a type signature: **no layer may leak a record**. It is exactly the condition under which composition preserves the interference the algorithm depends on.

The compiled-oracle caveat — read hardware claims carefully. Building $c-U_{a^{2^i}}$ honestly means the arithmetic stack of §13. Nearly every published “ $N = 15$ factored on hardware” demonstration instead uses a *compiled* oracle: knowing r in advance, one writes down the permutation $y \mapsto a^{2^i} y \bmod N$ as an explicit small circuit of NOTs, CNOTs and Fredkin gates, and notes that for $i \geq 2$ the permutation is the identity so no gate is needed at all. For $N = 15$, $a = 7$ this reduces the entire modular exponentiation to a handful of gates. This is a legitimate simulation aid and an illegitimate factoring algorithm: *the circuit is constructed from the answer*. It says nothing about whether a machine can run Shor’s algorithm on an input whose factorisation is unknown. When a cryptographic-risk assessment cites record factorisations as evidence about timelines, the first question to ask is whether the oracle was compiled — and the honest resource figures are the ones in §13, not the demonstration’s gate count.

Complexity, for the record

The dominant cost is the modular exponentiation, not the transform. Exponentiation by repeated squaring uses $O(n)$ modular multiplications of n -bit numbers, each costing $O(n \log n \log \log n)$ elementary operations by Schönhage–Strassen; the Hadamard layer costs $O(n)$ gates and the QFT $O(n^2)$ exactly, or $O(n \log n)$ to inverse-polynomial accuracy. A straightforward accounting gives $O(n^4)$ overall, and Shor’s tight bound is

$$O(n^2 \log n \log \log n) \text{ quantum steps,} \quad (67)$$

against $\exp(O((n \log n)^{1/2}))$ for the best known classical factoring algorithms — superpolynomial versus polynomial, which is the entire point.

14 What generalises, and what does not

- **Abelian HSP is solved.** One-dimensional irreps, an efficiently implementable QFT (§2–§5), uniform annihilator sampling (§6). Order finding, discrete log over $\mathbb{F}_p^\times$ and over elliptic-curve groups, and Simon’s problem are all instances. This is the exact cryptanalytic reach of Shor: RSA, finite-field Diffie–Hellman, DSA, ECDH and ECDSA fall together — one theorem applied five times, the shared premise being an *abelian* group with hidden periodicity.
- **Nonabelian HSP is not.** With irreps of dimension > 1 the Fourier transform is matrix-valued, measurement returns a representation label plus matrix indices, and reconstructing H appears to need entangled measurements across many coset states. Graph isomorphism sits at the symmetric group; *lattice problems reduce to the dihedral group*, where the best known algorithm is subexponential, not polynomial.

This is the structural answer to “why are the post-quantum replacements believed safe when RSA is not?” — the hardness of lattice problems does not present as an abelian hidden subgroup, and the one representation-theoretic route known runs into the dihedral case, open since the late 1990s. Note what this does *and does not* claim: the belief rests on the absence of an algorithm, not on a proof.

Migration consequences

- **ECC is not the safer half of an inventory.** Elliptic curves give equivalent classical security at far smaller parameters (256-bit ECC $\approx$ 3072-bit RSA), and that compactness — the reason ECC is everywhere in TLS, SSH and code signing — becomes a liability quantumly, since cost scales in the bit-length of the group. Any plan that prioritises RSA-2048 over P-256 has the order backwards.
- **Grover is not Shor.** Symmetric primitives lose at most half their security, so AES-256 and SHA-384 survive. Saying it out loud makes the asymmetric urgency more credible, not less.
- **The timing argument does not depend on a date.** Migration takes years, while harvest-now-decrypt-later means anything with a long confidentiality lifetime is already exposed. That is what defeats “wait and see” without needing to predict when a cryptographically relevant machine arrives.
- **Resource estimates are moving targets.** The counts above are logical-qubit and Toffoli figures under specific error-correction assumptions and improve with every optimisation paper. Quote them with their provenance, and re-check before putting numbers on a slide.

Appendix: one-page summary of the chain

	order finding	discrete log
group G_{HSP}	$\mathbb{Z}_Q$, rank 1	$\mathbb{Z}_N \times \mathbb{Z}_N$, rank 2
hidden subgroup	$r\mathbb{Z}$ — a <i>period</i>	$\langle (1, -\log_g x) \rangle$ — a <i>line</i>
secret is	the length of a cycle	the slope of a line
$ G $ known?	no — r is the secret	yes — public
QFT over	$\mathbb{Z}_{2^t}$, the wrong group	$\mathbb{Z}_N$, the right group
Lemma 2 used	approximately	exactly
output	$y/Q \approx k/r$, blurred	$(\tau \log_g x, \tau)$, sharp
classical stage	continued fractions, lcm, gcd	one modular inversion
success per run	$\varphi(r)/r$, plus $4/\pi^2$	$\varphi(N)/N$; $1 - 1/r$ if r prime

The chain, end to end

Order finding: $f(x+r) = f(x) \Rightarrow U_a^r = \mathbb{I}$ on the orbit $\Rightarrow \theta_k = \frac{k}{r} \xrightarrow{c-U_a^{2^m}} e^{2\pi i x k/r} \xrightarrow{F_Q^\dagger} y \approx Q \frac{k}{r} \xrightarrow{\text{CF}} r$.

Discrete log: $x = g^\lambda \Rightarrow f(\alpha, \beta) = g^{\alpha\lambda + \beta}$ hides $\langle (1, -\lambda) \rangle$
 $\xrightarrow{F_{\mathbb{Z}_N}^{\otimes 2}} (\tau\lambda, \tau)$ uniform on $H^\perp \xrightarrow{\tau^{-1}} \lambda$.

The underlying duality is classical Fourier analysis: repetition in position is localisation in frequency. What is quantum is the *access route*. Coherence lets one register hold a phase ramp across exponentially many values, written by only logarithmically many controlled group operations; orthogonality of the hiding function's values keeps distinct cosets from corrupting one another; interference in the inverse transform performs the read-out; and collapse converts the surviving correlation into one classical integer. Nothing in either circuit ever computes the secret — the circuit *changes basis* until the secret is the label of a measurement outcome.